

TOBERMORY, ON, Canada

WMO: 717670

Lat: 45.2239N

Lon: 81.6350W

Elev: 700

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.1	1.5	-8.7	3.5	-1.6	-4.5	4.4	2.9	23.7	23.9	20.9	23.7	7.1	0	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	78.0	67.4	75.7	66.5	73.6	65.7	70.1	74.9	68.8	73.2	67.5	71.6	7.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	106.6	72.7	67.1	102.3	71.6	65.8	97.7	70.3	34.5	74.9	33.4	73.4	32.3	71.8	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.5	15.6	DB	-9.3	82.5	7.5	2.9	-14.7	84.6	-19.2	86.2	-23.4	87.9	-28.9	89.9	(4)
(5)			WB	-9.7	72.6	7.5	1.9	-15.1	74.0	-19.5	75.1	-23.7	76.2	-29.2	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	20.9	20.6	27.7	38.5	50.3	58.7	65.2	65.2	59.0	48.4	37.0	28.3
	DBStd	17.71	10.60	9.96	9.89	7.52	7.00	5.38	4.59	4.49	6.50	7.29	7.88	8.10
	HDD50	4054	903	823	693	352	83	4	0	0	7	121	395	673
	HDD65	8020	1368	1243	1156	796	458	197	55	54	199	515	841	1138
	CDD50	1662	0	0	2	7	92	266	472	471	277	71	4	0
	CDD65	150	0	0	0	0	2	9	61	60	18	0	0	0
	CDH74	405	0	0	0	0	12	37	194	137	25	0	0	0
	CDH80	27	0	0	0	0	2	3	16	6	0	0	0	0
Wind	WSAvg	7.7	9.1	8.7	8.2	8.3	7.1	6.3	6.1	6.1	6.3	8.1	8.9	9.3
Precipitation	PrecAvg	39.0	2.9	2.1	2.2	3.2	3.5	3.3	3.3	3.7	3.9	4.3	3.5	3.0
	PrecMax	46.1	4.7	4.4	5.5	5.8	6.1	5.6	5.4	6.6	7.6	7.0	5.4	6.1
	PrecMin	31.9	1.5	0.8	0.3	1.2	1.3	1.1	1.5	1.9	1.5	1.9	1.6	1.3
	PrecStd	3.8	0.8	0.9	1.1	1.1	1.2	1.1	1.1	1.1	1.4	1.4	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	42.0	57.5	64.7	74.5	78.0	82.1	80.7	76.8	69.1	57.5	50.4
		MCWB	43.7	40.4	54.3	53.7	62.7	63.7	68.6	67.4	68.4	62.5	53.1	48.7
	2%	DB	40.1	38.7	49.1	58.2	70.0	73.9	78.5	77.4	73.9	65.8	54.0	44.4
		MCWB	38.9	37.2	44.1	48.7	59.2	62.9	67.5	68.1	67.6	61.1	50.4	42.8
	5%	DB	37.4	36.0	43.9	53.9	66.3	71.0	76.0	75.2	71.6	63.1	51.3	41.4
		MCWB	35.9	34.3	39.5	46.1	57.2	61.1	66.3	67.2	66.0	59.3	48.2	39.7
	10%	DB	35.2	33.6	40.3	50.0	62.5	68.4	73.6	73.1	69.2	60.2	49.0	39.2
		MCWB	33.5	31.8	36.7	43.6	54.6	59.7	65.1	66.2	64.0	56.4	46.2	37.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	41.0	54.0	56.1	65.1	67.6	72.0	71.6	70.5	65.4	54.0	49.3
		MCDB	44.0	41.3	57.6	61.8	72.6	73.5	78.0	76.8	74.2	67.7	56.8	50.4
	2%	WB	39.2	37.6	45.2	51.2	62.0	65.3	69.9	70.1	68.9	62.1	51.1	43.1
		MCDB	39.5	38.6	47.3	56.2	67.4	70.7	75.2	74.8	72.6	64.6	53.7	44.2
	5%	WB	36.1	34.4	40.3	47.5	58.8	63.3	68.2	69.0	67.3	59.5	48.6	40.0
		MCDB	37.1	35.6	43.3	52.1	63.6	68.1	73.6	73.2	70.6	62.4	50.9	41.2
	10%	WB	33.7	32.2	36.9	43.9	55.9	61.6	66.6	67.8	65.4	56.6	46.3	37.4
		MCDB	35.0	33.3	39.9	49.0	60.8	66.3	71.5	71.5	68.5	59.8	48.8	39.0
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.2	12.2	13.2	16.0	15.7	14.0	13.1	13.2	11.2	10.0	9.2
		MCDBR	11.5	11.7	14.4	20.1	20.2	19.0	16.9	15.0	14.2	14.1	12.5	10.6
		MCWBR	11.3	11.8	10.5	13.3	12.2	10.6	8.6	8.4	9.0	10.4	10.5	10.5
	5% WB	MCDBR	11.5	11.5	13.4	17.2	17.6	15.2	14.1	12.2	12.3	13.0	11.7	10.6
		MCWBR	11.6	11.9	11.2	13.8	12.7	10.4	8.6	8.2	9.1	11.0	10.7	10.7
Clear-Sky Solar Irradiance	taub	0.290	0.298	0.302	0.343	0.365	0.394	0.400	0.395	0.369	0.353	0.317	0.282	
	taud	2.472	2.396	2.440	2.369	2.423	2.358	2.381	2.386	2.437	2.442	2.496	2.533	
	Ebn at Noon	264	282	297	291	287	277	274	271	270	258	250	257	
	Edh at Noon	22	28	31	36	36	38	37	36	31	27	21	19	
All-Sky Solar Radiation	RadAvg	358	640	1101	1470	1768	1932	1959	1664	1276	749	403	259	
	RadStd	43	61	110	131	167	138	86	84	119	84	65	27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page